

Experiment No. 5

Write the Python program for Vacuum Cleaner Problem

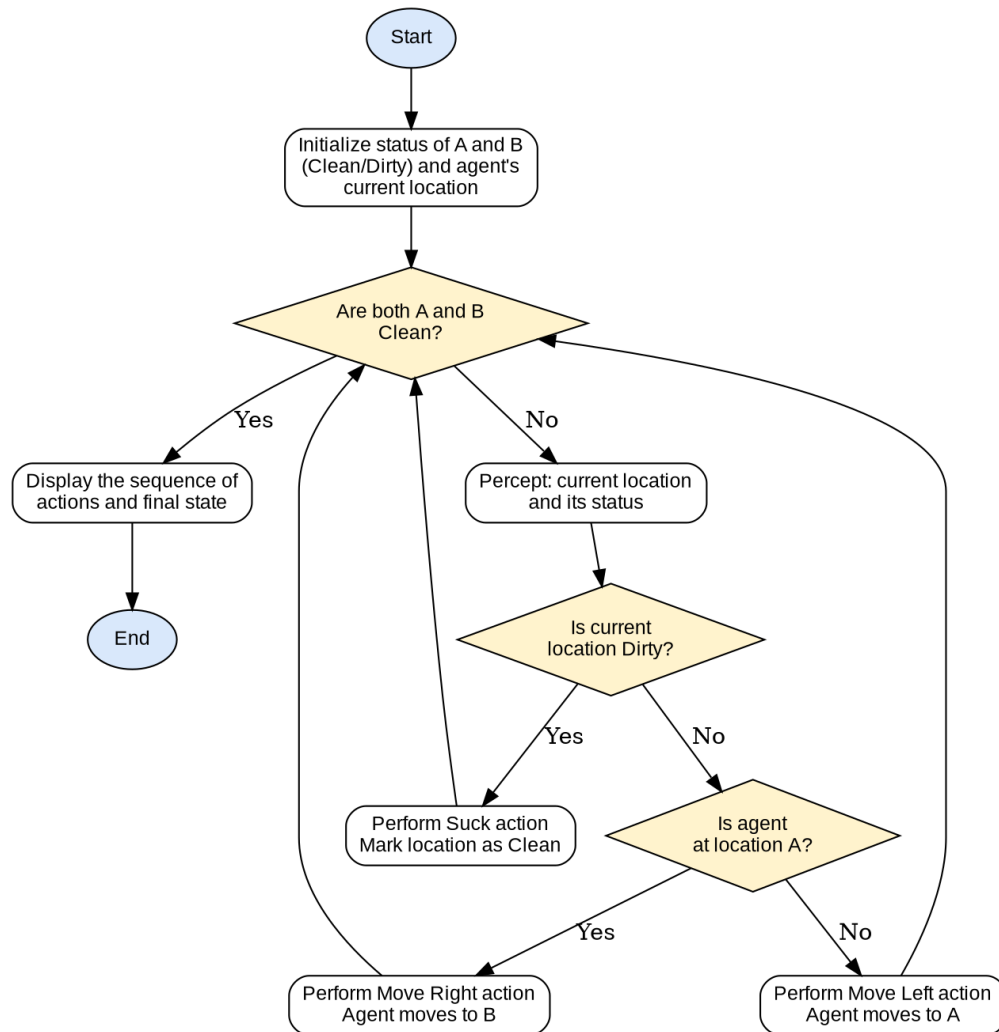
Aim

To write a Python program to implement a simple reflex vacuum cleaner agent that cleans a two-location environment (locations A and B) by sensing whether the current location is dirty and acting accordingly until both locations are clean.

Algorithm

1. Start.
2. Define the environment as two locations, A and B, each having a status of either Clean or Dirty, along with the agent's current location.
3. Initialize the status of A and B (e.g., both Dirty) and set the agent's starting location.
4. Repeat while at least one of the locations is Dirty:
5. (a) Percept: sense the agent's current location and its status.
6. (b) If the current location is Dirty, perform the Suck action and mark that location as Clean.
7. (c) Otherwise, if the agent is at location A, perform Move Right and move the agent to B.
8. (d) Otherwise (the agent is at location B), perform Move Left and move the agent to A.
9. When both A and B become Clean, stop the loop.
10. Display the sequence of actions performed and the final state of the environment.
11. Stop.

Flowchart



Program

```

def vacuum_cleaner(status, location):
    actions = []
    while status['A'] == 'Dirty' or status['B'] == 'Dirty':
        if status[location] == 'Dirty':
            actions.append(f"Suck at {location}")
            status[location] = 'Clean'
        elif location == 'A':
            actions.append("Move Right to B")
            location = 'B'
        elif location == 'B':
            actions.append("Move Left to A")
            location = 'A'
    return actions, status

```

```

status = {'A': 'Dirty', 'B': 'Dirty'}
location = 'A'

actions, final_status = vacuum_cleaner(status, location)

print("Simple Reflex Vacuum Cleaner Agent")
print("Initial state: A=Dirty, B=Dirty, Agent starts at A")
print("Actions performed:")
for i, action in enumerate(actions, start=1):
    print(f"{i}. {action}")
print("Final state:", final_status)

```

Result

The program was executed successfully and the output was verified as correct.

```

Simple Reflex Vacuum Cleaner Agent
Initial state: A=Dirty, B=Dirty, Agent starts at A
Actions performed:
1. Suck at A
2. Move Right to B
3. Suck at B
Final state: {'A': 'Clean', 'B': 'Clean'}

```

Sample Output — Running in VS Code

Screenshot of the program being run in the Visual Studio Code integrated terminal:

The image shows a Visual Studio Code editor window titled "vacuum_cleaner.py - AI-Lab - Visual Studio Code". The Explorer sidebar on the left shows a project named "AI-Lab" containing several files: "vacuum_cleaner.py", "bfs.py", "dfs.py", "tsp.py", "astar.py", and "map_coloring.py". The "vacuum_cleaner.py" file is open in the editor, displaying the following Python code:

```
1 def vacuum_cleaner(status, location):
2     actions = []
3     while status['A'] == 'Dirty' or status['B'] == 'Dirty':
4         if status[location] == 'Dirty':
5             actions.append(f"Suck at {location}")
6             status[location] = 'Clean'
7         elif location == 'A':
8             actions.append("Move Right to B")
9             location = 'B'
10        elif location == 'B':
11            actions.append("Move Left to A")
12            location = 'A'
13        return actions, status
14
15
16 status = {'A': 'Dirty', 'B': 'Dirty'}
17 location = 'A'
18
19 actions, final_status = vacuum_cleaner(status, location)
20
21 print("Simple Reflex Vacuum Cleaner Agent")
22 print("Initial state: A=Dirty, B=Dirty, Agent starts at A")
23 print("Actions performed:")
24 for i, action in enumerate(actions, start=1):
25     : (code continues — see Program section)
```

Below the code editor, the TERMINAL panel shows the output of running the script:

```
PS C:/Users/Athavan/AI-Lab> python vacuum_cleaner.py
Simple Reflex Vacuum Cleaner Agent
Initial state: A=Dirty, B=Dirty, Agent starts at A
Actions performed:
1. Suck at A
2. Move Right to B
3. Suck at B
Final state: {'A': 'Clean', 'B': 'Clean'}
PS C:/Users/Athavan/AI-Lab>
```

The status bar at the bottom indicates the file is "main", the cursor is at "Ln 1, Col 1", and the encoding is "Python UTF-8 LF Spaces: 4".